Project: Worked examples
Title: Bolts and fillet welds (EN 1993-1-8)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The design resistances of one bolt in a plate (shear, bearing, tension, punching, combined shear and tension), an optional block-tearing check of the bolt group, the spacing limits, and a fillet weld by the simplified and the directional method, each against its design force.
Bolt strengths: f_ub = class × 100 MPa; $A_s = \pi/4\cdot((d_2 + d_3)/2)^2$, $d_2 = d - 0.6495P$, $d_3 = d - 1.2269P$ Shear (cl. 3.6, Table 3.4): $F_{v,Rd} = \alpha_v \cdot f_{ub} \cdot A \cdot n_s/\gamma_{M2}$; $\alpha_v = 0.6$ (4.6, 5.6, 8.8 or shank in the plane), 0.5 (4.8, 5.8, 6.8, 10.9 with threads in the plane) Bearing: $F_{b,Rd} = k_1 \cdot \alpha_b \cdot f_u \cdot d \cdot t/\gamma_{M2}$; $\alpha_b = \min(\alpha_d, f_{ub}/f_u, 1)$, $\alpha_d = e_1/3d_0$ (end) or $p_1/3d_0 - 1/4$ (inner); $k_1 = \min(2.8e_2/d_0 - 1.7, 1.4p_2/d_0 - 1.7, 2.5)$ Tension: $F_{t,Rd} = k_2 \cdot f_{ub} \cdot A_s/\gamma_{M2}$, $k_2 = 0.9$; punching $B_{p,Rd} = 0.6\pi \cdot d_m \cdot t \cdot f_u/\gamma_{M2}$; combined $F_v/F_{v,Rd} + F_t/(1.4F_{t,Rd}) \le 1$ Block tearing (cl. 3.10.2): $V_{eff,1,Rd} = f_u \cdot A_{nt}/\gamma_{M2} + f_y \cdot A_{nv}/(\sqrt{3} \cdot \gamma_{M0})$ Weld, simplified (cl. 4.5.3.3): $f_{vw,d} = f_u/(\sqrt{3} \cdot \beta_w \cdot \gamma_{M2})$, $F_{w,Rd} = f_{vw,d} \cdot a$; directional (cl. 4.5.3.2): $\sqrt{(\sigma_{\perp}^2 + 3(\tau_{\perp}^2 + \tau_{\parallel}^2))} \le f_u/(\beta_w \cdot \gamma_{M2})$, $\sigma_{\perp} \le 0.9f_u/\gamma_{M2}$
Ref: EN 1993-1-8:2005 cl. 3.5, Table 3.3 — Positioning of holes for bolts
Ref: EN 1993-1-8:2005 cl. 3.6.1, Table 3.4 — Design resistance of individual fasteners
Ref: EN 1993-1-8:2005 cl. 3.10.2 — Block tearing
Ref: EN 1993-1-8:2005 cl. 4.5.3.2, 4.5.3.3, Table 4.1 — Fillet welds: directional and simplified methods; correlation factor
Ref: ISO 898-1:2013 cl. 9.1 — Nominal stress area (formula)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
grade	Bolt property class	8.8		drawing: SPL-001 rev B	any
size	Bolt size	M20		drawing: SPL-001 rev B	any
n_s	Shear planes	2		drawing: SPL-001 rev B	1 … 4
threads	Threads in the shear plane	yes		assumption: Threads may lie in the shear planes (short bolts).	any
steel	Plate steel grade (β_w)	S275		drawing: SPL-001 rev B	any
t	Plate thickness	10	mm	drawing: SPL-001 rev B	3 … 50 mm
f_y	Plate yield strength	275	MPa	spec: EN 10025-2 S275, t ≤ 16 mm	235 … 460 MPa
f_u	Plate ultimate strength	410	MPa	spec: EN 10025-2 S275, 3 ≤ t ≤ 100 mm	340 … 720 MPa
e_1	End distance (load direction)	40	mm	drawing: SPL-001 rev B	10 … 500 mm
e_2	Edge distance (across the load)	35	mm	drawing: SPL-001 rev B	10 … 500 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
p_1	Pitch (load direction; blank: one row)	70	mm	drawing: SPL-001 rev B	20 … 1000 mm
k_e1	Minimum e_1/d_o	1.2		spec: EN 1993-1-8 Table 3.3	1 … 3
k_e2	Minimum e_2/d_o	1.2		spec: EN 1993-1-8 Table 3.3	1 … 3
k_p1	Minimum p_1/d_o	2.2		spec: EN 1993-1-8 Table 3.3	2 … 4
k_p2	Minimum p_2/d_o	2.4		spec: EN 1993-1-8 Table 3.3	2 … 4
gamma_M2	γ_M2	1.25		spec: UK NA to BS EN 1993-1-8, NA.2.3	1.1 … 1.5
gamma_M0	γ_M0	1		spec: UK NA to BS EN 1993-1-1, NA.2.15	1 … 1.2
F_v_Ed	Shear per bolt	55	kN	assumption: Governing bolt from the bolt-group calculation (M3).	0 … 2000 kN
F_t_Ed	Tension per bolt	0	kN	assumption: No tension.	0 … 2000 kN

3. Assumptions

A1. Static loading; category A (bearing type) shear and category D (non-preloaded) tension; no slip-resistant joints or moment end plates. (Scope of S4.)
A2. Bearing is taken at the worst of an end bolt and an inner bolt, and at an edge bolt across the load (k_1 with e_2 and p_2). (Conservative for every bolt in the group.)
A3. Spacing minima (Table 3.3) and γ_M factors are entered from a National Annex record; the values shown default to EN 1993-1-8 recommended. (§A5 small factor set.)
A4. Group forces come from M3 and weld forces per length from M6 by link. (§A6 S4 links.)
A5. threads (Threads in the shear plane) is taken as yes. (Threads may lie in the shear planes (short bolts).)
A6. F_v_Ed (Shear per bolt) is taken as 55 kN. (Governing bolt from the bolt-group calculation (M3).)
A7. F_t_Ed (Tension per bolt) is taken as 0 kN. (No tension.)

4. Working

[19] f_ub = class first figure × 100 — EN 1993-1-8 Table 3.1
= class 8.8
= 800 MPa
Bolt ultimate strength

[20] $f_{yb} = f_{ub} \times$ second figure / 10 — EN 1993-1-8 Table 3.1
= class 8.8
= 640 MPa
Bolt yield strength

[21] $A_s = \pi/4 \cdot ((d_2 + d_3)/2)^2$, $d_2 = d - 0.6495P$, $d_3 = d - 1.2269P$ — ISO 898-1 cl. 9.1
= d = 20, P = 2.5 mm
= 244.79 mm²
Tensile stress area

[22] d_0 = d + 2 (≤ M24), d + 3 (≥ M27) unless entered
= 22 mm
= 22 mm
Hole diameter
[23] alpha_v = □_v (threads in the shear plane) EN 1993-1-8 Table 3.4
= class 8.8
= 0.6
Shear factor
[24] F_v_Rd = □_v·f_ub·A·n_s/□_M2 EN 1993-1-8 Table 3.4
= 0.6 × 800 × 244.79 × 2 / 1.25
= 188 kN
Shear resistance per bolt
[25] alpha_b = min(□_d, f_ub/f_u, 1), □_d = min($e_1/3d_0$, $p_1/3d_0$ - 1/4) EN 1993-1-8 Table 3.4
= min(0.60606, 0.81061, 1.9512, 1)
= 0.60606
Bearing factor α_b
[26] k_1 = min($2.8e_2/d_0$ - 1.7, $1.4p_2/d_0$ - 1.7, 2.5) EN 1993-1-8 Table 3.4
= e_2 = 35, p_2 = —, d_0 = 22
= 2.5
Bearing factor k_1
[27] F_b_Rd = k_1·□_b·f_u·d·t/□_M2 EN 1993-1-8 Table 3.4
= 2.5 × 0.60606 × 410 × 20 × 10 / 1.25
= 99.394 kN
Bearing resistance per bolt
[28] F_t_Rd = k_2·f_ub·A_s/□_M2, k_2 = 0.9 EN 1993-1-8 Table 3.4
= 0.9 × 800 × 244.79 / 1.25
= 141 kN
Tension resistance per bolt
[29] B_p_Rd = 0.6π·d_m·t·f_u/□_M2, d_m = (s + e)/2 EN 1993-1-8 Table 3.4
= 0.6π × 32.321 × 10 × 410 / 1.25
= 199.83 kN
Punching shear resistance
[30] u_v = F_v_Ed / F_v_Rd
= 55 kN / 188 kN
= 0.29255
Shear utilisation
[31] u_b = F_v_Ed / F_b_Rd
= 55 kN / 99.394 kN
= 0.55335
Bearing utilisation
[32] u_t = F_t_Ed / min(F_t_Rd, B_p_Rd)
= 0 kN / min(141 kN, 199.83 kN)
= 0
Tension utilisation (tension or punching)
[33] u_vt = F_v_Ed / F_v_Rd + F_t_Ed / (1.4 * F_t_Rd) EN 1993-1-8 Table 3.4
= 55 kN / 188 kN + 0 kN / (1.4 × 141 kN)
= 0.29255
Combined shear and tension

5. Results

SYMBOL	VALUE	DESCRIPTION
f_ub	800 MPa	Bolt ultimate strength
A_s	244.79 mm²	Stress area

SYMBOL	VALUE	DESCRIPTION
alpha_v	0.6	Shear factor
F_v_Rd	188 kN	Shear resistance
alpha_b	0.60606	Bearing factor
k_1	2.5	Bearing factor k_1
F_b_Rd	99.394 kN	Bearing resistance
F_t_Rd	141 kN	Tension resistance
B_p_Rd	199.83 kN	Punching resistance
u_v	0.29255	Shear utilisation
u_b	0.55335	Bearing utilisation
u_t	0	Tension utilisation
u_vt	0.29255	Combined utilisation
governing	"bearing (0.553)"	Governing bolt mode

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Bolt bearing	u_b = 0.55335	u_b ≤ 1	0.553	EN 1993-1-8 Table 3.4
PASS	Bolt shear	u_v = 0.29255	u_v ≤ 1	0.293	EN 1993-1-8 Table 3.4
PASS	Combined shear and tension	u_vt = 0.29255	u_vt ≤ 1	0.293	EN 1993-1-8 Table 3.4
PASS	Bolt tension and punching	u_t = 0	u_t ≤ 1	0	EN 1993-1-8 Table 3.4
PASS	Spacing and edge distances (Table 3.3 minima)	e_1 40, e_2 35, p_1 ≥ 26.4, 26.4, p_2 — mm 70, 48.4, 52.8 mm		0	EN 1993-1-8 Table 3.3

7. Applicability

WARNING — record.gamma_M2: gamma_M2: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.gamma_M0: gamma_M0: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.k_e1: k_e1: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.k_e2: k_e2: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.k_p1: k_p1: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.k_p2: k_p2: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.f_y: f_y: a tabulated value not taken from a reference record (Plate strengths); enter it as one so its source, edition and licence travel with it.
WARNING — record.f_u: f_u: a tabulated value not taken from a reference record (Plate strengths); enter it as one so its source, edition and licence travel with it.

8. Verification

Template struct.connection-ec3@1.0.0 Content hash 6ebee5888ba5fb09 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0